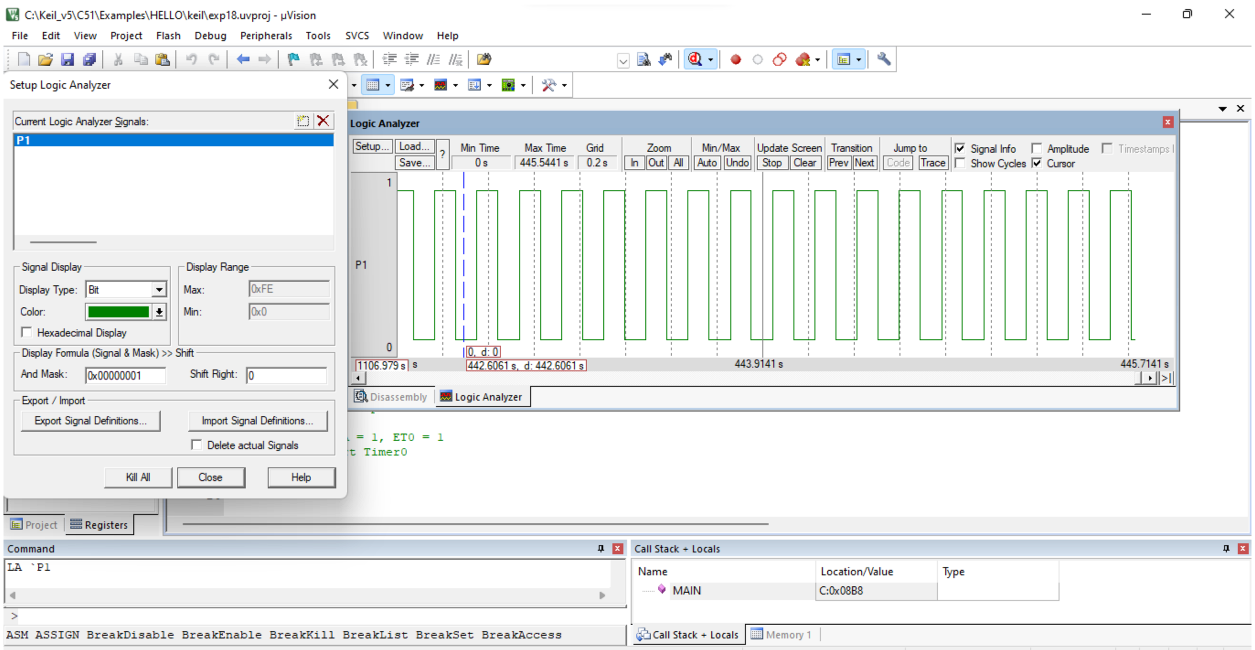
Embedded systems exp 18

Program:

```
#include <reg51.h>
sbit PWM = P1^0;
unsigned char count = 0;
unsigned char duty = 100; // Change this value (0–200)
void timer0_ISR(void) interrupt 1
{
 TH0 = 0xFC; // Reload for 1ms delay
 TL0 = 0x66;
 count++;
 if(count < duty)
 PWM = 1; // ON time
 else
 PWM = 0; // OFF time
 if(count >= 200) // Total period = 200ms
 count = 0;
}
void main()
{
 TMOD = 0x01; // Timer0 Mode 1 (16-bit)
 TH0 = 0xFC; // 1ms preload
 TL0 = 0x66;
 IE = 0x82; // EA = 1, ET0 = 1
 TR0 = 1; // Start Timer0
 while(1);
}
```

OUTPUT



C:\Keil_v5\C51\Examples\HELLO\keil\exp18.uvproj - µVision
File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Setup Logic Analyzer
Current Logic Analyzer Signals:
P1
Signal Display
Display Type: Bit
Color:
Hexadecimal Display
Display Range
Max: 0xFE
Min: 0x0
Display Formula (Signal & Mask) >> Shift
And Mask: 0x00000001
Shift Right: 0
Export / Import
Export Signal Definitions...
Import Signal Definitions...
Delete actual Signals
Kill All
Close
Help
Logic Analyzer
Setup... Load... Save...
Min Time 0 s
Max Time 445.5441 s
Grid 0.2 s
Zoom In Out All
Min/Max Auto Undo
Update Screen Stop Clear
Transition Prev Next
Jump to Code Trace
Signal Info
Show Cycles
Amplitude
Cursor
1
P1
0
1106.979 s
0, d: 0
442.6061 s, d: 442.6061 s
443.9141 s
445.7141 s
Disassembly
Logic Analyzer
= 1, ET0 = 1
t Timer0
Project
Registers
Command
LA `P1
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess
Call Stack + Locals
Name
Location/Value
Type
MAIN
C:0x08B8
Call Stack + Locals
Memory 1